MIL-C-82178(MC)
13 February 1968
SUPERSEDING
See Section 6.

MILITARY SPECIFICATION

CLOTH, GABARDINE, WOOL: 12 OUNCE

1. SCOPE

1.1 This specification covers the requirements for one type of wool gabardine cloth.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

CCC-T-191 - Textile Test Methods.

PPP-P-51 - Packaging, Packing And Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks And Synthetics).

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections.

MILITARY

MIL-STD-105 - Sampling Procedures And Tables For Inspection By Attributes.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U.S. DEPARTMENT OF AGRICULTURE

Methods of Test for Grades of Wool Top

(Application for copies should be addressed to U. S. Department of Agriculture, Agricultural Marketing Service, Washington, D. C. 20380.)

Rules and Regulations under the
Wool Products Labeling Act of 1939.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Price may be obtained from the Superintendent of Documents.)

FSC 8305

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3. REQUIREMENTS

3.1 Standard sample. The cloth shall match the standard sample for shade and finish and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.2 Preproduction sample approval. Unless otherwise specified (see 6.2) before production is commenced, a 3-yard full width sample of the finished cloth shall be submitted or made available to the contracting officer or his authorized representative for approval. The approval of the preproduction sample authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of the specification. The preproduction sample shall be manufactured by the supplier in the same facilities to be used for the manufacture of the production items.

3.3 Material.

3.3.1 Wool stock. The wool top shall be made from fleece wool, pulled wool or any combination thereof not lower in grade than 70's U. S. Standard (see 2.2). The wool shall be of suitable staple length to meet the requirements of the specification. The use of top made from laps, noils or any other manufacturing by-product is prohibited.

3.3.2 Yarn. The yarn for the warp and filling shall be combed, drawn and evenly spun and shall have sufficient twist to meet the requirements of the specification. The warp and filling yarn shall be two ply.

3.4 Color. The color shall match an approved shade standard of blue 2321 or green 2231 as specified (see 6.2 and 6.4). The color shall be produced by blending dyed wool top with dyes that will produce the desired color, evenness of shade and meet the specified fastness requirements. The green color shall be produced by blending colored wools with white wools.

3.4.1 Matching. The color shall match the approved standard sample under natural (north sky) daylight, or artificial daylight, having a color temperature of 7500° Kelvin and shall be a good approximation to the approved standard sample under incandescent lamplight at 2800° Kelvin.

3.4.2 Colorfastness. The finished cloth shall show colorfastness to light, wet-dry cleaning, perspiration, and crocking equal to or better than the standard sample when tested as specified in 4.4. When no standard sample is established or not designated as applicable to colorfastness, the finished cloth shall show good fastness to light, wet-dry cleaning perspiration, laundering and crocking when tested as specified in 4.4.

3.5 Physical requirements. The finished cloth shall conform to the minimum requirements of Table I when tested as specified in 4.4.

Table I - Physical requirements

Weight linear yard, 56 in. basis ounces		Yarns per inch	Breaking strength pounds	Shrinkage Percent	Weave
Min.	Max.	Minimum	Minimum	Maximum	
12.0	13.0	Warp 128 Filling 66	Warp 120 Filling 50	Warp 4.0 Filling 3.0	2 2 right twill

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3.6 Width. The minimum width of the finished cloth, exclusive of the selvages, shall be 60 inches.

3.7 Finish. The cloth shall be thoroughly scoured, closely sheared and well pressed so as to provide a finish equal to that of the standard sample.

3.7.1 Moth repellent treatment. The cloth shall be given a moth repellent treatment in accordance with the method specified in 6.3. The pH of the finished cloth shall be not less than 3.3 nor more than 4.5 when tested as specified in 4.4.

3.8 Length and put-up. Unless otherwise specified (see 5.1 and 6.2), the cloth shall be furnished in continuous pieces of not less than 50 yards. Each length shall be put up in full width rolls as specified in PPP-P-51.

3.9 Fiber identification. Each roll of cloth shall be labeled, ticketed, or invoiced for fiber content in accordance with the Wool Products Labeling Act of 1939 (see 2.2).

3.10 Workmanship. The finished fabrics shall conform to the quality of products established by this specification. The demerit points per 100 square yards when calculated as specified in Section 4 shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificates of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certifications.

4.2 Inspection. Inspection shall be in accordance with MIL-STD-105 except where otherwise indicated.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

4.2.1.1 Wool. The wool shall be examined for grade in top form by comparison with the applicable U.S. Standard. In the event of a dispute, the wool grade shall be determined by the width method (wedge). The sample unit shall be 1 yard of top. The sample size (number of sample units) shall be as shown below. The lot size shall be expressed in units of 1 pound. The results from each sample unit shall be reported separately. The lot shall be unacceptable if a single unit fails to meet the required grade.

Lot Size (Pounds)	Sample Size
800 or less	2
801 up and including 22,000	3
22,001 and over	5

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4.2.1.2 In-process inspection. Inspection shall be made at any point during any phase of the manufacturing process to assure that no noils or laps and only approved wools, in the amounts specified, are used and that the dyeing requirements have been adhered to (see 3.4). The Government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated non-conformance.

4.2.2 Examination of the end item. Examination of the end item shall be in accordance with 4.2.2.1, 4.2.2.2, 4.2.2.3, and 4.2.2.4.

4.2.2.1 Yard-by-yard examination. The entire yardage of each roll in the sample, but not more than 100 yards per roll, shall be examined on the face side. The sample unit shall be 1 yard. The number of rolls from which the sample yardage shall be extracted shall be in accordance with the following:

Lot Size (Yards)	Sample Size (Rolls) 1/
3200 or less	10
3201 up to and including 10,000	15
10,001 and over	25

1/ No more than one roll shall be taken from any shipping container unless the number of shipping containers in the lot is less than the required number of rolls, in which case, all shipping containers shall be present in the sample.

All defects as defined in Section II of Fed. Standard No. 4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1.1. The same yardage shall be given a through-lighting inspection from the back of the cloth toward a light source for holes and thin areas not previously scored on the face side of the cloth; slight open places resulting from removal of burrs, shives or sisal not previously scored on the face side shall not be scored in through-lighting inspection. Defects on the selvage shall not be scored unless the body of the cloth is affected. No continuous linear yard or increments of 1 yard on the measuring device of the inspection machine from any one roll within the sample shall be penalized more than 4 points. For determination of acceptability on the average of the lot, the lot shall be unacceptable if the points per 100 square yards exceed 10.0 points. For determination of acceptability on the individual roll quality basis the lot shall be unacceptable if two or more individual rolls exceed 18.0 points per 100 square yards. If no individual roll exceeds this point level, the lot shall be accepted. If one roll exceeds this point level, a second sample, same size as first sample, shall be selected for individual roll quality examination only. If no roll in the second sample exceeds the point value, the lot shall be accepted. If one or more rolls exceed this value, the lot shall be rejected. For individual roll quality examination, local type defects only shall be scored. Overall type defects shall be scored only in determining compliance with the lot average quality requirements. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample size} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 square yards.}$$

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4.2.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	- one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

NOTE: The following defects, when present, shall be scored four points for each yard in which they occur.

Degree and character of finish on face and back not equal to standard sample.

Mottled, cloudy or streaky.

Width less than specified.

Baggy, ridgy or wavy cloth.

Edges of roll build up or softness of roll in body of cloth.

Shading defects - Shaded side to side, side to center, or end to end.

4.2.2.2 Examination for length.

4.2.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards below the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.2.2.2.2. Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on roll tickets. The rolls examined shall be those selected for the examination of individual rolls.

4.2.2.3 Examination for compliance with the wool products labeling act. During the yard-by-yard examination, each roll in the sample shall be examined for conformance to the Wool Products Labeling Act. Each roll not labeled in accordance with this act shall be a defect. The lot shall be unacceptable if two or more of these defects occur.

4.2.2.4 Examination for shade, individual rolls. Each roll in the lot shall be examined visually for shade match (see 3.4.1). A roll shall be unacceptable if it fails to meet the requirements for shade match. The sample unit shall be a 4-inch by 20-inch swatch of the cloth. A sample unit shall be drawn from each roll in the lot.

4.3 Examination of preparation for delivery requirements. Examination shall be made in accordance with the provisions of PPP-P-51 to determine that packaging, packing, and marking requirements of Section 5 of this specification are complied with.

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4.4 Tests. The methods of testing specified in CCC-T-191, wherever applicable and as listed in Table II shall be followed. The physical and chemical values specified in Section 3 apply to the average results of the determination made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 3 continuous yards, full width of the finished cloth. The lot size shall be expressed in units of 1 yard each. The lot shall be unacceptable if one or more sample units fail to meet any test requirements specified. All test reports shall contain the individual value utilized in expressing the final results. The sample size shall be in accordance with the following:

Lot Size (Yards)	Sample Size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE II TEST METHODS

Characteristic	Rqmt. para	Test Method
Fiber content	3.3.1	2101
Yarn ply 1/	3.3.2	visual
Weave 1/	Table I	visual
Weight, oz/lin yd	Table I	5041
Yarns per inch, warp	Table I	5050
filling	Table I	5050
Breaking strength, warp	Table I	5100
filling	Table I	5100
Shrinkage	Table I	5590
Colorfastness to:		
Light	3.4.2	5660
Wet-dry cleaning	3.4.2	5622
Perspiration	3.4.2	5682
Crocking	3.4.2	5650
Acidity (pH)	3.7.1	2811
Moth repellent treatment	3.7.1	2/

1/ One determination per sample unit and the results reported as "pass" or "fail".

2/ Certificate of compliance

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Levels A and C. The cloth shall be put up on rolls and packaged in accordance with the requirements for class 1 fabrics of PPP-P-51.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-51.

5.3 Marking. In addition to any special marking required by the contract or order, the shipping containers shall be marked in accordance with the requirements of PPP-P-51.

6. NOTES

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6.1 Intended use. The cloth is intended for use in the manufacture of uniform items worn by personnel of the U. S. Marine Corps.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of specification.
(b) Whether preproduction sample approval is required (3.2).
(c) Color required (3.4).
(d) Width required (3.6).
(e) Acceptance of short pieces (3.8 and 6.5).
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Method of moth repellent treatment. The cloth shall be moth repellent treated in accordance with the method described in the Marine Corps publication entitled "Silicofluoride Treatment for Woolens and Worsteds", dated 6 February 1953, revised 12 November 1957. Copies of the publication may be obtained upon request to the procuring activity issuing the invitation for bids.

6.4 Shade sample. For information regarding availability of standard shade specified, address inquiry to the procuring activity issuing the invitation for bids.

6.5 Short pieces. When specified (see 6.2), pieces of cloth under 50 yards in length, not to exceed 5 percent of the total contract yardage, will be accepted under the following conditions:

Length of pieces	Discount from contract price
40 to 49-7/8 yards	5 percent
30 to 39-7/8 yards	10 percent

Short pieces shall be prepared for delivery in the same manner as specified for full length pieces (see 5.1). Each roll shall contain only one continuous piece of cloth. Two or more short pieces of cloth joined or rolled together, regardless of total length, will not be accepted.

6.6 Preproduction sample. Examination, tests and approval shall be as specified by the contracting officer (see 3.2).

6.7 Supersession data. This specification supersedes Marine Corps Purchase Description PD-64 (MCSA) dated 4 November 1959 for Cloth, Wool, Gabardine; 12 Ounce.

Preparing activity:
Navy - MC
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